LEFT EXACTNESS VIA FINITE PRODUCTS, EQUALIZERS, AND FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3524-3534, following the statement at 3510-3522.

Editorial note. The equalizer identification is displayed data with a routine universal-property check recorded as an obligation; no omission markers in the span.

Allowed standard prerequisites (editorial).

- Left exactness equals commuting with all finite limits. the definition of left exact in force (finite limits context) and lemma-limits-products-equalizers, cited at proof line 3525. Supplies the implication (2) implies (1): a functor commuting with finite products and equalizers commutes with all finite limits.
- Existence of finite limits in A. lemma-finite-limits-exist, cited at statement line 3514. Fixes the standing hypothesis that all finite limits exist in A, so the displayed iterated fibre products exist.

Source reference. Original lemma 002O: sources/stacks-categories/pinned-original-source.tex, source lines 2265–2265. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 002N: sources/stacks-categories/pinned-original-source.tex, source lines 1745–1745. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be a functor. Suppose all finite limits exist in $\mathcal{A}$, see Lemma 002O. The following are equivalent:*

1. *F is left exact,*
2. *F commutes with finite products and equalizers, and*
3. *F transforms a final object of $\mathcal{A}$ into a final object of $\mathcal{B}$, and commutes with fibre products.*

Proof. Lemma 002N shows that (2) implies (1). Suppose (3) holds. The fibre product over the final object is the product. If $a, b : A \to B$ are morphisms of $\mathcal{A}$, then the equalizer of a, b is

$$(A \times_{a,B,b} A) \times_{(\mathrm{pr}_1,\mathrm{pr}_2),A\times A,\Delta} A.$$

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Thus (3) implies (2). Finally (1) implies (3) because the empty limit is a final object, and fibre products are limits. □